

MSPhysics: Complete Tools & Operations Guide

Comprehensive User & Technical Reference Manual for Real-Time Physics in Trimble SketchUp

Executive Overview: MSPhysics is an advanced real-time rigid-body and kinematic simulation suite embedded directly within Trimble SketchUp. Built atop the high-performance **Newton Dynamics 3.14** C++ physics engine and coupled with **AMS Library 3.8.0**, it enables architects, mechanical designers, animators, and engineers to simulate physical mechanisms, evaluate structural dynamics, model multi-body kinematics, drive vehicles, and export photorealistic keyframed physics animations. The suite is organized into distinct specialized toolsets covering live execution, joint articulation, transmission gearing, timeline replay, deep property inspection, and build-time cross-compilation.

Master Toolsets Taxonomy

Toolset Family	Count	Core Capabilities & Functional Focus	Primary Access Point
Simulation Controls	5 tools	Full model play, selective execution (Play 2), pause, reset, freeze-in-place stop, and UI inspector toggle.	Toolbar: <i>MSPhysics</i>
Joint Connection & Scaling	3 tools	Interactive joint-to-body binding tool with modifier keys (Ctrl/Shift), joint scaling gizmo, and gear pairing.	Toolbar: <i>MSPhysics Joints</i>
Joint Creation & Actuators	14 joints	Parametric kinematic constraints: Hinge, Motor, Servo, Slider, Piston, Spring, Corkscrew, Ball/Socket, Universal, Fixed, Plane, Curvy Slider, Curvy Piston, UpVector.	Toolbar: <i>MSPhysics Joints</i>
Replay & Keyframing	10 tools	Deterministic frame recording, camera tracking, bidirectional playback, speed scaling, scene page generation.	Toolbar: <i>MSPhysics Replay</i>
UI Dialog & Scripting	5 tabs	World gravity/solver settings, 10 collision mesh types, material densities, thrusters/emitters, Ruby IDE, 3D audio.	Dialog: <i>Toggle UI</i>
Developer & Build Tools	3 suites	Zig C++ cross-compiler for Ruby 3.2 UCRT, upstream C++ patcher, and headless WASM test harness.	Scripts: <i>tools/</i> & <i>harness/</i>

Essential 4-Step Physics Workflow

STEP 1: Model & Group	STEP 2: Place Joints	STEP 3: Connect & Bind	STEP 4: Simulate & Control
Encapsulate solid geometry into Groups or Component Instances. Assign collision shape (e.g. Box, Convex Hull) and material/mass in UI.	Select a joint tool (e.g. Hinge). Click Point 1 for anchor origin; click Point 2 to align primary constraint axis (Blue Z-vector).	Activate Joint Connection Tool. Click the Body, hold CTRL and click the Joint. Joint turns GREEN confirming active binding.	Click Toggle Play . Interact with dynamic bodies using mouse spring drag, control sliders, or keyboard controllers (WSADQE).

Core Physical Classifications & Solver Architecture

Dynamic Rigid Bodies: Top-level groups possessing mass and inertia, fully governed by forces, gravity, and Newton collision laws.	Static Colliders: Top-level geometry marked as 'Static' or untouched in selective simulation; acts as immovable world terrain.
Kinematic / Frozen Bodies: Bodies controlled explicitly by scripts or joints; unaffected by external forces until awakened.	Substepping & Continuous Collision (CCD): Solver advances at 60–240 Hz with CCD to prevent fast objects from tunneling thin walls.

Physics Units, Coordinate Systems & Numerical Modeling Standards

Physical Quantity	Standard Unit	Default Constant / Value	Engineering Modeling Best Practice
Length & Distance	Meters (m) or Inches (in)	SketchUp internal unit (inches converted to m)	Optimal rigid-body scale is 0.1m to 50m. Scale up tiny millimeter parts.
Mass & Density	Kilograms (kg), g/cm³	Default material density: 1.0 g/cm³ (Water)	Avoid mass ratios > 100:1 between directly connected joints to prevent jitter.
Force & Torque	Newtons (N), N·m	Thrusters, springs, joint limits	Dynamic controllers evaluate in Newtons: $F = m \cdot a$.
Coordinate Frame	Right-handed Z-Up	Red = +X, Green = +Y, Blue = +Z	Global Gravity points along world -Z vector by default ([0, 0, -9.8] m/s²).

SECTION 1: SIMULATION TOOLBAR & INTERACTIVE CONTROLS

CORE EXECUTION

The **MSPhysics Simulation Toolbar** is the primary command hub for running, testing, pausing, and freezing dynamic physical simulations. Unlike standard animations, MSPhysics executes a true non-linear numerical differential solver in real time. Below is the complete reference for every toolbar command, live interactive viewport controls, and solver execution mechanics.

Tool & Icon	Command / State	Primary Physical Function	Step-by-Step Operational Guide	Shortcut / Prompt
 Toggle UI	Modal / Modeless State: Checked / Unchecked	Opens or dismisses the floating MSPhysics Inspector Dialog containing Simulation, Body, Joint, Script, and Sound settings.	1. Click button to open inspector. 2. Select any body/joint in SketchUp. 3. Dialog dynamically updates fields to match active entity.	Status: <i>Show/hide MSPhysics UI</i>
 Toggle Play	Global Sim Engine State: Play / Pause	Initiates real-time physics simulation across the entire model. Every top-level group/component becomes an active dynamic body.	1. Verify model geometry is grouped. 2. Click button to launch simulation. 3. Click again mid-sim to freeze/pause solver; click to resume.	Status: <i>Play/Pause simulation (all groups)</i>
 Reset	Reversion Tool State: Active during Sim	Terminates simulation, restores all moved bodies back to their exact pre-simulation transformations, and destroys emitted dynamic copies.	1. Click at any time during or after paused simulation. 2. Physics engine clears memory and resets model state.	Status: <i>End simulation and reset positions</i>
 Toggle Play 2 (Selective)	Selective Engine State: Selection Active	Starts simulation ONLY from highlighted groups/components. All non-selected entities are automatically treated as stationary/static obstacles.	1. Use SketchUp Select tool to highlight specific bodies. 2. Click Toggle Play 2. 3. Only selection moves; prevents lagging in giant scenes.	Status: <i>Play/Pause from selected groups</i>
 Stop (Freeze Poses)	State Baker Tool State: Active during Sim	Halts simulation WITHOUT restoring initial positions and WITHOUT deleting emitted bodies. Bakes final physics poses into the SketchUp model.	1. Run simulation until objects settle naturally (e.g. falling rocks, dumped gravel, draped ropes). 2. Click Stop to keep final layout permanently.	Status: <i>End simulation keeping positions</i>

Live In-Simulation Interactive Tools & Viewport Manipulation

 Interactive Mouse Drag Tool	 6-DOF Flight & Walk Camera
While simulation is running, Left-Click and Drag any dynamic body in 3D viewport. A virtual spring attaches from cursor hit point to body surface, applying real physical pulling force. Holding SHIFT while dragging applies rotational torque, allowing interactive orientation tilting.	Navigate viewport freely during active physics without interrupting solver: • W / S : Fly forward / backward A / D : Strafe left / right • Q / E : Elevate up / down Arrow Keys : Pan / tilt camera look-at angle • CTRL + Left/Right Arrow : Roll camera clockwise / counter-clockwise.
 On-Screen Control Panel Sliders	 Right-Click Context Menu Tools
When any joint or thruster controller uses slider(name, default, min, max), MSPhysics generates an on-screen HUD window with graphical sliders. Users drag sliders in real-time to steer wheels, articulate robotic arms, or control thrusters.	Right-clicking any group/component in SketchUp provides instant access to MSPhysics context submenus: • State : Toggle Static, Frozen, Ignored, Collidable, Auto Sleep, Continuous Collision, Gravity, Magnetic. • Shape : Override collision primitive Material : Assign friction presets.

Simulation Engine Solver Loop & Performance Tuning

Phase 1: Input & Event Ingestion	Phase 2: Controller & Force Eval	Phase 3: Newton Substepping	Phase 4: Viewport Synchronization
AMS Library polls Windows message loop, intercepting async keystrokes, mouse spring coordinates, and UI slider adjustments.	Evaluates all Ruby controller expressions, thruster force vectors, and magnetic field equations before solver integration.	Newton Dynamics executes iterative LCP constraint solver (60–240 Hz), resolving joint kinematics, friction, and CCD contacts.	Transforms calculated by C++ engine are pushed into SketchUp Group transformations; camera positions updated.
Solver Stability & Optimization Pro-Tips: • Substepping: If high-speed mechanisms explode or jitter, increase <i>Update Rate</i> to 120Hz or 240Hz in the Simulation UI tab. • Continuous Collision Detection (CCD): Always enable CCD for fast-moving small projectiles or thin wall boundaries to prevent tunneling. • Auto Sleep: Keep Auto Sleep enabled so resting bodies deactivate solver calculations, freeing CPU cycles for moving bodies.			

SECTION 2: JOINT CONNECTION TOOL & GEAR TRANSMISSIONS

KINEMATIC BINDING

In MSPhysics, **joint placement** is strictly decoupled from **joint connection**. Simply inserting a joint component next to a body does *not* constrain it. The **Joint Connection Tool** establishes the mathematical parent-child linkage between rigid bodies and constraint frames. Furthermore, the **Gear Connection Tool** couples multiple joints with transmission gear ratios for mechanical engineering.

The Joint Connection Tool & Modifier Key Architecture

Modifier Key	Cursor Graphic	Action Mode	Operational Procedure & Mouse Interaction	Viewport Visual Indicator
None (Default)		Select Entity	Click on any body (group/component) or joint to make it active selected entity. Prepares entity for connection.	Solid Blue Outline: Selected entity in memory.
CTRL		Connect (+)	While a body is selected, hold CTRL and click a Joint to bind them. Alternatively, select a joint and Ctrl-click child bodies.	Solid Green Outline: Connected joint or child body.
SHIFT		Disconnect (-)	While a body is selected, hold SHIFT and click an already connected joint to sever kinematic constraint linkage.	Red Selection Cue: Connection severed.
CTRL + SHIFT		Toggle (+/-)	Hold both keys simultaneously and click any target joint to invert connection state (connects if severed; severs if connected).	Adaptive Toggle: Swaps active connection state.

Component Multi-Instance Propagation: When connecting a joint to a SketchUp Component Instance that possesses multiple copies in the model, MSPhysics detects the shared ComponentDefinition and displays a dialog: *Would you like to connect/disconnect this joint to/from all alike instances?*

- Clicking **YES** replicates the joint constraint across every identical instance automatically (e.g. four vehicle wheels).
- Clicking **NO** establishes a unique connection for only the clicked instance.

The Gear Connection Tool & Transmission Kinematics

Transmission Theory & Ratio Formula	Step-by-Step Gear Pairing Workflow
Gears mathematically couple degrees of freedom between two separate joints: Rotational to Rotational: $Speed2 = -Ratio * Speed1$ (Spur gears, planetary sets) Rotational to Linear: $Velocity2 = Ratio * Omega1$ (Rack & pinion, lead screws) Linear to Linear: $Velocity2 = Ratio * Velocity1$ (Scissor linkages) Supported Joints: Hinge, Motor, Servo, Slider, Piston, Spring, CurvySlider, CurvyPiston.	- Ensure both bodies are top-level and each is connected to at least one supported joint. - Activate the Gear Connection Tool. - Click instance A (turns Blue); hold CTRL/SHIFT and click instance B (turns Green). - Open MSPhysics UI -> Joint Tab -> Gears section. - Enter desired Gear Ratio (positive for matching; negative for counter-rotation).

Joint Connection Rules & Mechanical Archetypes Matrix

Mechanism Type	Joint Combination	Gear Ratio Formula	Engineering Setup & Practical Guidance
Spur Gear Reduction	Hinge A + Hinge B	$Ratio = -(Teeth\ A / Teeth\ B)$	Negative sign forces counter-rotation. Place hinges at exact pitch circle centers.
Rack and Pinion	Hinge A + Slider B	$Ratio = Radius\ (meters)$	Couples rotating pinion gear to linearly translating rack rail. Used for steering.
Mechanical Scissor Lift	Slider A + Slider B	$Ratio = -1.0$	Symmetrically couples left and right sliding pins, ensuring balanced vertical lift.
Compound Differential	Motor + 2 Hinges	$Omega_M = 0.5 * (W_L + W_R)$	Allows drive wheels to rotate at differing speeds during vehicle cornering.

Joint Connection Integrity Rules: (1) *Connecting to Self:* A body cannot be connected to itself (raises warning). (2) *Interconnecting Bodies:* Directly connecting Body to Body without an intermediate joint is prohibited. (3) *Ignored Bodies:* Connecting to an ignored body or a joint inside an ignored body is disallowed. (4) *Articulated Chains:* To create robotic arms or excavators, nest the joint inside the parent body group; connect child body to it.

SECTION 3: JOINT CREATION TOOLS — ROTATIONAL & LINEAR ACTUATORS

ACTUATORS & RAILS

MSPhysics features 14 specialized parametric joint types. Every joint is placed via a 2-point vector sequence: **Point 1 sets the joint's anchor origin; Point 2 defines its primary constraint direction** (represented by the Blue axis). Exact dimensions can be entered in SketchUp's Measurements (VCB) box. Below is the complete engineering specification for the 7 primary rotational and linear actuator joints.

Joint & Icon	DOFs & Type	Mechanical Applications	Key Parameters & Limits	Controller Formula & Expressions
 Hinge	1 DOF: Rotation around Blue Z-axis	Doors, cabinet flaps, wheels, pendulums, robotic elbows, casters, levers.	- Min/Max angle limits (-180° to +180°) - Angular friction & damping - Spring stiffness & rest angle	Motorized hinge controller: slider('hinge', 0, -90, 90)
 Motor	1 DOF: Powered continuous rotation	Propellers, car wheels, cooling fan turbines, saw blades, conveyor rollers.	- Target speed (rad/s or deg/s) - Acceleration & max torque - Free-wheeling when controller is 0	Dynamic speed controller: 15.0 * (key('w') - key('s'))
 Servo	1 DOF: Target angle positioner	Steering knuckles, aircraft ailerons/rudders, robotic arms, pan/tilt gimbals.	- Angular sweep limits - Max angular rate (deg/s) - Power/holding torque limit	Target angle in degrees: 35 * (key('right') - key('left'))
 Slider	1 DOF: Linear passive translation	Drawers, sliding glass doors, crane gantries, elevator guide tracks.	- Linear Min/Max limits (meters) - Linear friction coefficient - Linear spring stiffness & damping	Passive constraint by default. Acts as spring buffer when spring parameters > 0.
 Piston	1 DOF: Powered linear actuator	Hydraulic/pneumatic rams, dump truck lifters, elevator pistons, jacks.	- Stroke length limits (Min/Max) - Linear travel speed (m/s) - Push/pull max force (Newtons)	Target position along stroke: slider('lift', 0, 0, 2.5)
 Spring	1 DOF: Elastic linear or torsional motion	Vehicle suspension struts, pogo sticks, trampoline springs, recoil buffers.	- Stiffness k (N/m) - Damping c (N·s/m) - Rest distance & travel bounds	Hooke's Law: $F = -k(x - x_0) - c \cdot v$. Tuned in Joint UI tab.
 Corkscrew	2 Coupled DOFs: Simultaneous slide & twist	Threaded bolt screws, spiral augers, spiral ramps, screw bottle caps.	- Pitch (travel distance per 360° turn) - Linear bounds & angular limits - Rotational & linear friction	Coupled motion constraint: $\Delta Z = \text{Pitch} * (\Delta \theta / 2\pi)$.

Joint Coordinate Axes Convention: Each joint visual gizmo displays colored Cartesian axes. **Blue represents the primary constraint axis** (the axis of rotation for Hinge/Motor/Servo; the axis of slide for Slider/Piston). **Red and Green represent constraint limits and zero-reference planes.** Ensure the Blue axis is precisely aligned with your intended axle or guide track before connecting.

Dynamic Controller Expressions Syntax & Functions

Expression Syntax Pattern	Input Context	Mathematical Behavior	Typical Robotic / Mechanical Use Case
slider('name', def, min, max)	Control Panel HUD	Exposes on-screen real-time graphical slider returning float between min and max.	Manual steering, crane boom angle, valve throttling, variable speed.
val * (key('w') - key('s'))	Keyboard Polling	Returns +val on 'w', -val on 's', and 0.0 when neither or both are pressed.	Forward / reverse vehicle drive, hoist lift, bidirectional motors.
amp * sin(2 * PI * freq * time)	Oscillator Math	Generates continuous harmonic sinusoidal oscillation as time advances.	Windshield wipers, pendulum clock escapement, walking gait animators.
joystick(:left_y) * 100	Game Controller	Reads analog joystick axis value (-1.0 to +1.0) with proportional sensitivity.	Precision flight simulator yoke, analog gamepad throttle and steering.

Articulated Kinematic Chains & Hierarchy Guidelines: To build multi-segment robot arms, backhoes, or folding landing gear, do not leave all joints at top-level world space. Instead, group Joint 2 inside Body 1's group; then connect Body 2 to Joint 2. This embeds Joint 2's frame of reference into Body 1's moving local coordinate space, creating robust compound kinematic linkages.

SECTION 4: JOINT CREATION TOOLS — SPATIAL CONSTRAINTS & PATHS				SPATIAL & PATH CONSTRAINTS
Complex mechanisms require multi-dimensional spatial constraints, spherical pivots, planar glides, or path-following trajectories. MSPhysics includes sophisticated joint geometries capable of binding bodies along arbitrary 3D curves or stabilizing multi-axis orientations. This section covers the remaining 7 joint types plus the Joint Scaling tool.				
Joint & Icon	DOFs & Type	Mechanical Applications	Key Parameters & Settings	Setup Guide & Placement Notes
 BallAndSocket	3 DOFs: Spherical omnidirectional rotation	Automotive tie-rods, suspension wishbones, anatomical shoulders/hips, trailer hitches.	• Max cone cone-angle limit • Axial twist limits (Min/Max) • Joint friction & damping	Place center at the pivot sphere origin. Aligns freely in 3D space.
 Universal	2 DOFs: Orthogonal angular rotation	Cardan drive shaft couplings, mechanical gyroscope rings, wrist articulations.	• Min/Max angle for Axis 1 • Min/Max angle for Axis 2 • Rotational friction	Place at the cross-pivot intersection of the two rotational yokes.
 Fixed	0 DOFs: Rigid structural weld	Rigidly locking two dynamic bodies together, or welding a body immovably to world space.	• Breakable max linear force (N) • Breakable max torque (N·m) • Rigid lock when breakable is 0	Place anywhere on contact plane. Snaps under extreme impact if breakable is set.
 Plane	3 DOFs: 2 Linear (X/Y) + 1 Rotational (Z)	Air hockey pucks, hovercraft, furniture gliding on flat floors, planar XY pantographs.	• Linear surface friction • Angular surface friction • Strict normal axis constraint	Align Blue Z-axis perpendicular to the sliding surface plane.
 CurvySlider	1 Path DOF: Passive sliding along 3D spline	Rollercoasters, overhead factory conveyor rails, train tracks, curtain tracks.	• Loop path mode (cyclic) • Linear friction along curve • Alignment: Lock orientation to tangent	1. Group a continuous chain of edges. 2. Insert CurvySlider at curve origin. 3. Connect body with Connection Tool.
 CurvyPiston	1 Path DOF: Powered drive along 3D spline	Automated cable cars, camera motion dollies along splines, automated guided vehicles (AGV).	• Target position along path • Speed & drive power limit • Loop mode & curve alignment	Controller drives distance along curve length: slider('track_pos', 0, 0, 50)
 UpVector	Orientation Lock: Restricts pitch and roll	Self-balancing rovers, speedboats, hovercraft, upright walking biped characters.	• Max correction torque • Angular damping • Target world vector ([0, 0, 1])	Prevents dynamic bodies from tipping or flipping over during aggressive motions.

Scale MSPhysics Joints Tool & Distance Constraints

 Scale MSPhysics Joints Tool	Point-to-Point & Cable Constraints
In large architectural models or miniature mechanical watches, joint gizmos may appear too small or too large. Activating Scale MSPhysics Joints displays an input prompt to rescale joint visual geometry (from 0.05x to 10.0x). Crucial Invariance Note: Rescaling joints modifies <i>only</i> visual viewport representation. It does not change physical mass, stiffness, damping, limits, or solver behavior.	PointToPoint (Joint ID 15): A spherical tether constraint that maintains a fixed distance between two anchor points in space while allowing unrestricted 3-axis angular freedom. Used for suspension chains, crane hoisting cables, bridge stays, and flexible fabric links. Configurable with breaking tension limits.

Curvy Spline Construction & Structural Failure Mechanics

Curvy Spline Edge Chain Construction	Breakable Constraints & Crash Destruction
1. Create a continuous chain of connected SketchUp Edges or Arcs. 2. Group the curve geometry. Curvy joints read vertex sequences within the group. 3. Insert CurvySlider or CurvyPiston at the initial starting vertex. 4. In UI, check <i>Loop</i> for closed circuits, or set tangent alignment to prevent car derailment.	Joints can be configured to simulate catastrophic structural failure under excessive shock: • Breakable Force (N): Snaps joint if linear tension/shear exceeds threshold. • Breakable Torque (N·m): Snaps joint if bending moment exceeds threshold. Essential for bridge collapse tests, breakaway bolts, and crumple zones.
Vehicle Chassis & Suspension Assembly Pattern: A typical 4-wheel vehicle uses a main Chassis Body. At each wheel position, insert a vertical Spring joint (suspension travel). Attach a steering knuckle body to the spring bottom with a Servo joint (steering angle). Finally, connect the wheel to the servo knuckle with a Motor joint (drive torque). Add an UpVector to the chassis if necessary to eliminate rollover on extreme slopes.	

SECTION 5: MSPHYSICS UI DIALOG & INSPECTOR TOOLS			INSPECTION & SCRIPTING
The MSPhysics UI Dialog (launched via Toggle UI) is an interactive, multi-tabbed web-based inspector. It communicates bidirectionally with SketchUp's Ruby core to configure world physics, body properties, joint parameters, custom programmatic behaviors, thruster/emitter engines, and 3D spatial audio.			
Inspector Tab	Target Domain	Key Configurable Properties & Settings	Operational Instructions & Best Practices
Tab 1: Simulation	World Environment & Solver Settings	• Gravity Vector: Global X, Y, Z acceleration (default: -9.8 m/s² along Z) • Continuous Collision (CCD): Eliminates high-speed tunneling • Update Rate & Substeps: 60Hz, 120Hz, 240Hz solver frequency • Visual Debuggers: Collision Wireframe, Center of Mass axes, Bounding Boxes, Contact Points & Force vectors.	Enable Collision Wireframe to inspect how Newton Dynamics sees your geometry. Increase Update Rate to 120/240Hz for stiff mechanisms or fast vehicles.
Tab 2: Properties	Body Mesh, Mass, Thrusters & Emitters	• 10 Collision Shapes: Box, Sphere, Cone, Cylinder, Capsule, Chamfer Cylinder, Convex Hull, Compound, Compound from CD, Null • Mass Control: By Mass (kg) or by Density (g/cm³) • Material Properties: Static/Kinetic Friction, Elasticity, Softness • Magnetic Force & Range: Attract, Repel, or Dipole modes • Thruster Engine: Directional force vector with controller • Emitter Cannon: Spawns copies of body at rate, speed, lifetime.	Always prefer Primitives (Box, Cylinder) or Convex Hull over complex meshes for massive performance gains. Use Compound from CD for assemblies with internal holes.
Tab 3: Joint	Joint Limits & Controllers	• Kinematic Limits: Min/Max angles, stroke distances • Controllers: Dynamic mathematical formulas and key bindings • Friction & Springs: Angular/linear damping and stiffness • Breakable Thresholds: Force and torque break limits.	Select any joint in the model to reveal its dedicated tuning panel. Write expressions like 1000 * key('space') into the Controller field.
Tab 4: Script	Ruby Scripting IDE & Event Callbacks	Embedded Ace Code Editor with Ruby syntax highlighting. Lifecycle Event Hooks: • onStart { ... }: Runs once upon simulation start • onUpdate { ... }: Runs every frame before physics step • onPostUpdate { ... }: Runs after physics integration • onTouch(contact) { ... }: Triggered on collision • onUntouch(contact) { ... }: Triggered when contact ends • onDestroy { ... }: Cleanup when simulation terminates.	Use full access to MSPhysics Body API: body.set_velocity(vec), body.add_force(f), body.set_color(c). Script errors are cleanly reported in console.
Tab 5: Sound	SDL2 Mixer Audio System	• Supports WAV, MP3, OGG, FLAC, AIFF, MIDI • 3D Positional Audio: Spatial falloff, stereo panning, Doppler effect • Volume, pitch, and distance attenuation controls.	Trigger sounds on collision events or controller thresholds: simulation.play_sound('boom.wav', body.get_position).

Thruster Vector Engine & Emitter Projectile System

Thruster Engine Configuration	Emitter Projectile Cannon System
Thrusters apply dynamic force vectors to their associated bodies in real time: • Controller Input: Number in Newtons (e.g. 500) or 3D vector [Fx, Fy, Fz]. • Lock to Body Axes: When enabled, force rotates along with body (jet engines, rockets). When disabled, force acts in global world directions. • Steering: Combine with keys: [1000*key('w'), 0, 500*key('space')].	Emitters dynamically spawn clones of their associated body during simulation: • Rate & Delay: Spawn frequency interval (seconds) and initial countdown. • Lifetime: Lifespan before emitted copies are automatically erased (0 = permanent). • Recoil: Implements Newton's 3rd Law—applies opposite reaction impulse to the cannon body upon projectile launch.

Ruby Scripting API Quick Reference (Body & Simulation Methods)

Ruby API Method	Target Object	Function & Parameter Types	Example Usage & Context
body.get_velocity / set_velocity(v)	MSPhysics::Body	Gets or overrides linear velocity vector [vx, vy, vz] in m/s.	body.set_velocity([0, 0, 10]) (Instant vertical jump)
body.add_force(f) / add_torque(t)	MSPhysics::Body	Applies instantaneous force (Newtons) or rotational torque (N-m).	body.add_force([0, 500, 0]) inside onUpdate hook
simulation.play_sound(path, pos)	MSPhysics::Simulation	Plays 3D spatial audio file at given Point3d in model space.	simulation.play_sound('hit.wav', contact.point)
simulation.find_body_by_name(n)	MSPhysics::Simulation	Finds active simulation Body instance by its SketchUp entity name.	target = simulation.find_body_by_name('Turret')

SECTION 6: REPLAY & KEYFRAME ANIMATION TOOLBAR

ANIMATION & EXPORT

The **MSPhysics Replay Toolbar** is an enterprise-grade animation recording and playback suite. During live physics simulation, millions of calculations occur per second. Replay records every body transformation, velocity vector, and viewport camera movement into memory and persistent model attributes, enabling deterministic playback, camera choreography, and photorealistic raytraced rendering.

Tool & Icon	Command Name	Engine Function & Behavior	Operational Usage & Workflow	Impact & Notes
 Toggle Record	Record Armed (Toggle)	Enables recording mode. When active, every simulation frame captures body positions, orientations, and visibility states.	1. Click button (turns Checked). 2. Press Toggle Play to run simulation. 3. Replay data streams into buffer.	Captures full rigid body kinematics.
 Camera Replay	Camera Tracking (Toggle)	Synchronously records SketchUp viewport camera eye, target, and up-vector during simulation alongside bodies.	1. Enable before pressing Play. 2. Orbit/pan camera during physics run. 3. Camera path replays automatically.	Enables cinematic camera fly-throughs.
 Play	Forward Playback (Action)	Plays recorded simulation timeline forward from current frame to the end of the captured sequence.	1. Reset simulation after recording. 2. Click Replay Play to review motion.	Smooth interpolation at native FPS.
 Reverse	Backward Playback (Action)	Plays recorded animation backward in reverse time toward frame 0.	1. Click during or after replay. 2. Useful for rewind effects and impact analysis.	Exact reverse kinematic trajectory.
 Pause	Freeze Frame (Toggle)	Temporarily halts replay playback at the current frame without resetting timeline position.	1. Click to pause. 2. Click Play or Reverse to resume.	Allows viewport inspection of any frame.
 Reset	Rewind to 0 (Action)	Stops replay playback and rewinds all model geometry and camera back to initial Frame 0.	Click to return model to starting positions.	Prepares timeline for re-playback.
 Stop	Halt at Frame (Action)	Terminates replay playback but keeps objects at their current frame transforms (avoids resetting to 0).	Click when desired pose is reached.	Bakes replay frame into model geometry.
 Speed Adjust	Increase / Decrease (Step commands)	Dynamically scales playback speed multiplier from 0.01x (super slow-motion) up to 10.0x (high-speed fast-forward).	Click Increase Speed or Decrease Speed . Status bar displays current multiplier.	Real-time timescale modulation.
 Clear Data	Purge Memory (Destructive)	Permanently erases all cached replay frame data from RAM and SketchUp model attributes.	1. Click when animation is no longer needed. 2. Frees RAM and prevents large file size.	Reclaims storage space in model file.

Photorealistic Rendering & Video Export Workflow:

1. **Export to Native SketchUp Scenes (Pages):** In MSPhysics Replay UI, click 'Export to Scenes' to convert keyframes into native SketchUp animation tabs.

2. **Raytraced Rendering:** Compatible with all major SketchUp render engines: **V-Ray, Enscape, Thea Render, Twilight Render, and Kerkythea**. The renderer advances through the scene tabs, generating photorealistic raytraced frames with full motion blur and global illumination.

3. **Image Sequence / MP4 Export:** Export uncompressed PNG/JPEG sequences to compile broadcast-quality video animations.

Production Export Pipelines & Memory Architecture			
Replay Memory Budget & Data Serialization • RAM Buffer: Replay stores transformations in efficient C++ native memory buffers during recording to prevent garbage-collector hitches. • Save to Model: Commits frame records as serial binary attributes into the .skp model, allowing animation to persist across sessions. • Save to File: Streams massive simulations directly to external .mspr replay files to avoid bloating model file size.		FFmpeg High-Definition Video CLI Pipeline To composite raw frame images into standard 60 FPS ProRes / MP4 video: ffmpeg -framerate 60 -i frame_%05d.png -c:v libx264 -preset slow -crf 18 -pix_fmt yuv420p output.mp4 • Preserves exact frame cadence without frame drops. • Lossless color fidelity matching SketchUp viewport outputs.	
Rendering Engine	Integration Method	Motion Blur & Lighting	Recommended Settings
Chaos V-Ray	Batch Scene Tab Rendering	Physical camera motion blur enabled	Disable scene transition delays; set frame interval to 1.0 frame.
Enscape 3D	Camera Path Video Sync	Real-time raytracing, screen space reflections	Lock time of day; enable Replay Camera tracking.
Thea / Twilight Render	Animation Timeline Exporter	Spectral unbiased path tracing	Export camera and geometry matrices; utilize GPU denoising.

SECTION 7: DEVELOPER, BUILD & DIAGNOSTIC TOOLS			BUILD CHAIN & TROUBLESHOOTING
This repository contains the complete cross-compilation toolchain and automated testing harness that updated MSPhysics to run natively on SketchUp 2024–2026 (Ruby 3.2 UCRT) . Understanding these developer tools enables engineers to inspect native engine bindings, rebuild binary extensions, and diagnose runtime faults.			
Tool / Script Path	System Role	Technical Mechanism & Pipeline	Execution Command & Requirements
tools/prepare_sources.sh	Source Fetcher & Patch Applier	Downloads upstream Newton Dynamics 3.14 and SDL C++ sources; applies non-MSVC patches (ruby_prep.h.patch, dgTypes.cpp.patch) for GCC/Clang/Zig compatibility.	tools/prepare_sources.sh [dest] Requires: curl, tar, patch
tools/build_win64_ruby32.sh	Zig C++ Cross-Compiler	Cross-compiles native extensions msp_lib.so and newton.dll targeting Windows x64 Ruby 3.2 (x64-ucrt-ruby320). Uses Zig's Clang frontend to target x86_64-windows-gnu without requiring Microsoft Visual Studio.	tools/build_win64_ruby32.sh Requires: pip install ziglang
native/harness/run_tests.sh	Ruby WASM Test Harness	Executes test suite on Ruby 3.2 inside WebAssembly (ruby.wasm) with a stubbed SketchUp 2026 API. Validates syntax compilation, extension loading, AMS fallback routines, and DLL import dependencies.	native/harness/run_tests.sh Requires: node >= 18, @ruby/3.2-wasm-wasi

Built-in Diagnostic & Troubleshooting Systems

Loader Hardening & ABI Verification	PE Import Table Inspector (load_error.txt)
MSPhysics/main_entry.rb intercepts extension loading. It verifies that AMS Library >= 3.8.0 is present and that a matching native binary exists for host Ruby ABI before invoking C++ libraries. Prevents silent SketchUp crashes and reports human-readable error dialogues.	If a DLL fails to load, MSPhysics directly parses the Portable Executable (PE) import table of msp_lib.so. It checks every dependency (newton.dll, SDL2.dll, SDL2_mixer.dll) and writes a comprehensive report to MSPhysics/load_error.txt.
AMS Library Self-Repair: If cleanup scripts inadvertently delete ruby_fallback.rb, MSPhysics automatically restores it from MSPhysics/ams_lib_fallback/ upon startup.	Ruby 3.2 Compatibility: Fully eliminates deprecated File.exists?, Fixnum, Bignum, and keyword argument splat syntax.

Master Keyboard & Operational Quick-Reference Matrix

Key / Input	Context / Active Tool	Triggered Operational Action	Visual / Physics Feedback
Spacebar / Click	Simulation Toolbar	Toggle Play / Pause simulation solver	Runs Newton dynamic integration / freezes time
Left-Click Drag	Live Simulation Viewport	Picks and throws dynamic rigid body	Applies elastic spring force vector to cursor hit
Shift + Drag	Live Simulation Viewport	Applies rotational torque to picked body	Tilts and twists body along angular cursor path
W, S, A, D, Q, E	Live Simulation Viewport	6-Axis flying camera navigation	Translates camera forward, backward, strafe, elevate
Arrow Keys	Live Simulation Viewport	Camera pan, tilt, and look-around	Rotates camera viewpoint angles
Ctrl + Click	Joint Connection Tool	Binds selected body to clicked joint	Target joint highlights in Solid Green
Shift + Click	Joint Connection Tool	Severs connection between body and joint	Removes constraint linkage
Ctrl + Shift + Click	Joint Connection Tool	Toggles active connection state	Connects if severed; severs if connected
Top 5 Troubleshooting Solutions: (1) <i>Engine Fails to Load:</i> Verify Ruby 3.2 UCRT binary is installed at MSPhysics/libraries/stage/win64/3.2/msp_lib.so. (2) <i>AMS Library Error:</i> Ensure AMS Library 3.8.0+ is active. (3) <i>SketchUp 2026.0 Matrix Inversion:</i> If singular transforms occur, MSPhysics auto-reverts to identity matrix. (4) <i>ShadowInfo KeyError in 2026.1+:</i> Extension automatically skips protected shadow attributes. (5) <i>Tunneling:</i> Enable Continuous Collision Detection (CCD) in Simulation settings.			